Week 1 – Day 2 Deliverable

Healthcare IAM Operations Simulation

RBAC Design, Simulated Applications, Organizational Unit (OU) Structure, and Security Group Design

Project Name	Healthcare IAM Operations Simulation
Organization	Horizon Valley Medical Center (HVMC)
Project Week	Week 1
Day	Day 2
Document Version	1.0
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1. Purpose

The objective of Day 2 is to design the Identity and Access Management (IAM) foundation by identifying business applications, defining Role-Based Access Control (RBAC), creating a logical Organizational Unit (OU) structure, and establishing a standardized security group naming convention. These components will support user provisioning, access reviews, onboarding, transfers, and offboarding throughout the remainder of the project.

2. Simulated Business Applications

Application	Business Purpose	Departments Using the Application
Electronic Health Record (EHR)	Patient records and clinical documentation	Doctors, Nurses
Pharmacy Management System	Medication management	Pharmacy
Laboratory Information System (LIS)	Lab orders and results	Laboratory, Doctors
HR Portal	Employee records	Human Resources
Payroll System	Salary and payroll processing	HR, Finance
Finance ERP	Budgeting and financial reporting	Finance
IT Administration Console	User account administration	IT
Security Monitoring Portal	Physical security and incident monitoring	Security
Appointment Scheduling System	Patient appointment scheduling	Reception
VPN Portal	Remote access	IT, Contractors

3. Role-Based Access Control (RBAC) Matrix

Role	EHR	Pharmacy	LIS	Payroll	HR Portal	Finance ERP	IT Admin	Security Portal	Scheduling
Doctor	Read / Write	Read	Read	—	—	—	—	—	—
Nurse	Read / Write	Read	Read	—	—	—	—	—	—
Pharmacist	Read	Read / Write	—	—	—	—	—	—	—
Laboratory Technician	Read	—	Read / Write	—	—	—	—	—	—
Human Resources	—	—	—	Read / Write	Read / Write	—	—	—	—
Finance	—	—	—	Read / Write	Read	Read / Write	—	—	—
IT Administrator	Read	Read	Read	—	—	—	Admin	Admin	Admin
Security Officer	—	—	—	—	—	—	—	Admin	—
Receptionist	Read	—	—	—	—	—	—	—	Read / Write
Contractor	Limited	—	—	—	—	—	Limited	—	—

4. Organizational Unit (OU) Structure

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Horizon Valley Medical Center
│
├── Users
│   ├── Doctors
│   ├── Nursing
│   ├── Pharmacy
│   ├── Laboratory
│   ├── Human Resources
│   ├── Finance
│   ├── IT
│   ├── Security
│   ├── Reception
│   └── Contractors
│
├── Groups
│
├── Computers
│
├── Service Accounts
│
└── Disabled Users
```

5. Security Group Naming Convention

Naming Standard

SG-<Department>-<Application>-<Access>

Examples

Security Group	Purpose
SG-Doctors-EHR-RW	Doctors with Read/Write access to EHR
SG-Nursing-EHR-RW	Nurses with Read/Write access to EHR
SG-Pharmacy-PharmacySystem-RW	Pharmacy application access
SG-Lab-LIS-RW	Laboratory Information System access
SG-HR-HRPortal-RW	HR Portal access
SG-Finance-Payroll-RW	Payroll access
SG-IT-ITAdmin-Admin	IT administrative access
SG-Security-SecurityPortal-Admin	Security monitoring access
SG-Reception-Scheduling-RW	Appointment scheduling access
SG-Contractors-VPN-Read	Contractor VPN access

6. IAM Design Principles

The following principles will guide access management throughout the project:

- Role-Based Access Control (RBAC)
- Principle of Least Privilege
- Group-Based Permission Assignment
- Separation of Duties
- Centralized Identity Management
- Standardized Naming Conventions
- Audit Readiness

7. RBAC Governance Policies (Decide Your RBAC Rules)

To guide the remainder of the project, the following IAM policies have been documented. These rules establish how roles, access, and identity lifecycle events will be governed across Horizon Valley Medical Center.

- Every employee is assigned one primary role.
- Access is granted through security group membership, not directly to users.
- Users receive the minimum access required (Least Privilege).
- Contractors receive temporary accounts with limited permissions.
- IT Administrators approve all access requests.

- HR notifies IT when employees join, transfer, or leave.
- Departing employees are disabled on the same day.

8. Security Considerations

- Access is assigned to security groups, not individual users.
- Users receive permissions based on their job role.
- Contractors receive limited access.
- Administrative privileges are restricted to IT Administrators.
- Access changes will follow documented onboarding, transfer, and offboarding processes.

9. Key Outcomes

At the conclusion of Day 2:

- A standardized list of business applications was documented.
- An RBAC matrix was created to define role-based permissions.
- An Active Directory OU structure was designed.
- A security group naming convention was established.
- IAM governance principles and RBAC policies were documented to support future identity lifecycle activities.

10. Deliverables

- Simulated Application Inventory
- RBAC Matrix v1
- Organizational Unit (OU) Structure
- Security Group Naming Convention
- IAM Design Principles
- RBAC Governance Policies
- Security Considerations
- Day 2 Documentation for GitHub Portfolio